

ROSHAN KUMAR

Kanpur, Uttar Pradesh

+91 8081761956 | roshankumar89603@gmail.com

[linkedin.com/in/roshan-kumar-a835b4361](https://www.linkedin.com/in/roshan-kumar-a835b4361)

Git-Hub- [roshan-kumar2 \(Roshan Kumar\)](#)

Career Objective

Aspiring Software Developer and Data Science student with strong knowledge of C++, Python, and Data Structures. Skilled in Software Development, Backend Development, Data Analysis, and Problem Solving. Seeking an entry-level role to build scalable and efficient solutions.

Technical SKILLS

Languages: C, C++, Java, Python

Libraries/Tools: NumPy, Pandas, SUMO

Web Technologies: HTML, CSS, JavaScript

Core Concepts: Data Structures, Algorithms, OOP, Problem Solving

Technologies: IoT, V2I Communication, Real-time Systems

EDUCATION

B.Tech in Computer Science (Data Science)

Pranveer Singh Institute of Technology, Kanpur | 2023–2027

12th (ICSE)

Dr. Brij Kishori Dubey Memorial School, Kanpur | 2021–2022

EXPERIENCE/INTERNSHIP

Web Development Intern

Cognifyz Technologies | Feb 2026 – Mar 2026

- Developed responsive web pages using HTML, CSS, and JavaScript
- Improved UI/UX design for better user interaction
- Applied frontend development and debugging techniques

PROJECTS

Personal Voice Assistant | Python | Speech Recognition

2023-2024

- Built a voice assistant using **speech recognition and NLP techniques**
- Processed voice commands and automated basic tasks
- Improved user interaction using **real-time audio processing**

Smart Traffic Management System using V2I Technology

2026-2027

- Designed and developed a smart traffic management system using Vehicle-to-Infrastructure (V2I) communication to reduce urban traffic congestion.
- Implemented real-time data processing using Python for vehicle speed, position, and traffic flow analysis.
- Developed a Safe-to-Pass-First (SPF) algorithm for intelligent intersection control and collision avoidance.
- Applied backend logic and real-time data analysis for traffic optimization
- Integrated Dashboard Traffic Light (DBTL) sensors for seamless IoT-based communication between vehicles and infrastructure.
- Simulated and tested the system using SUMO (Simulation of Urban Mobility) for performance evaluation.
- Achieved 23% reduction in average waiting time, improving traffic throughput and system efficiency.
- Strengthened concepts of Intelligent Transportation Systems (ITS), IoT, real-time systems, and smart city technologies.

CERTIFICATIONS

- Oracle University – Data Science Professional Certification
- Infosys Springboard – Python, HTML/CSS

ACHIEVEMENTS & LEADERSHIP

- Solved 100+ problems on LeetCode (Data Structures & Algorithms)